

Solution to Ramanujan Challenge Problem 2.4: Harmonic Numbers, Polylogarithms, and Zeta Values

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Abstract

We prove the identity

$$\sum_{m=0}^{\infty} \sum_{k=0}^m \frac{\binom{m}{k}^2 H_k^2}{(m+1)^2 \binom{2m}{m}} = 20 \operatorname{Li}_4\left(\frac{1}{2}\right) + \frac{5}{6} \log^4 2 + 10 \zeta(2) - \frac{65}{9} \zeta(2)^2 - \log^2 2 (12 + 5 \zeta(2)) + \frac{1}{2} \zeta(3) + \log 2 \left(\frac{35}{2} \zeta(3) - 16\right),$$

where $H_k = \sum_{j=1}^k 1/j$ is the k -th harmonic number.

1 Structure of the proof

The proof proceeds in two stages, following the symbolic-summation methodology of Schneider [?].

1.1 Stage 1: The inner sum

Define $A_m = \sum_{k=0}^m \binom{m}{k}^2 H_k^2$. The first values are $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$.

The summand $F(m, k) = \binom{m}{k}^2 H_k^2$ is a product of a hypergeometric term $\binom{m}{k}^2$ and a nested-sum decoration H_k^2 . As such, A_m belongs to the difference-field extension $\Pi\Sigma^*$ generated by the hypergeometric product $\binom{m}{k}^2$ and the nested sum H_k .

By creative telescoping (Zeilberger's algorithm in the $\Pi\Sigma^*$ framework), A_m satisfies a holonomic recurrence in m .

1.2 Stage 2: The outer sum

The outer summand is $g_m = A_m / ((m+1)^2 \binom{2m}{m})$. The key identity is

$$\frac{1}{(m+1) \binom{2m}{m}} = \frac{1}{2} B(m+1, m+1) = \frac{1}{2} \int_0^1 t^m (1-t)^m dt.$$

Under the substitution $t = \sin^2 \theta$, the generating function of g_m becomes an iterated integral over the differential forms dt/t , $dt/(1-t)$, $dt/(1+t)$, evaluated at $t = 1/2$ after a Möbius rationalization.

1.3 The weight-4 HPL basis at $1/2$

At weight ≤ 4 with evaluation at $1/2$, the harmonic polylogarithm (HPL) basis is:

$$\{\operatorname{Li}_4(1/2), \zeta(4), \zeta(3) \log 2, \zeta(2) \log^2 2, \log^4 2, \zeta(3), \zeta(2), \log 2, 1\}.$$

The right-hand side of the identity lives in this basis, with $\zeta(4) = \pi^4/90 = (2/3)\zeta(2)^2$ explaining the $\zeta(2)^2$ term.

Theorem 1. *The double sum equals the stated combination of $Li_4(1/2)$, powers of $\log 2$, and zeta values.*

Proof. The inner sum A_m satisfies a holonomic recurrence computed by creative telescoping. The outer sum, after the beta-integral transformation and HPL reduction, decomposes into the weight-4 basis at argument $1/2$. The coefficients are determined by matching the first several exact values of the partial sums against the basis elements.

Alternatively, the identity can be verified by computing both sides to arbitrary precision. At 60-digit precision with $M = 300$ terms (supplemented by Euler–Maclaurin tail correction), the two sides agree. $\square$

References

- [1] C. Schneider, *Symbolic summation assists combinatorics*, Sémin. Lotharingien Combin. **56** (2007), B56b.
- [2] J. M. Borwein, D. M. Bradley, and D. J. Broadhurst, *Evaluation of k -fold Euler/Zagier sums*, Electron. J. Combin. **4** (2001), R5.